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(54) **T4 DNA LIGASE VARIANTS WITH INCREASED LIGATION EFFICIENCY**

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C12Q 1/686 (2018.01)

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CPC ***C12Q 1/6855*** (2013.01); ***C12Q 1/686*** (2013.01)

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CPC C12Q 1/6855; C12Q 1/686; C12Q 1/6862; C12Q 1/6806; C12Q 2521/501; C12N 9/93; C12N 15/70; C12P 19/34; C12Y 605/01001
See application file for complete search history.

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(57) **ABSTRACT**

The invention includes a mutant T4 DNA ligase or a biologically active fragment thereof, which has greater activity than wild type T4 DNA ligase. The mutant T4 DNA ligase, or the biologically active fragment, has one or more substitutions differing from the wild type, as described more fully in the Summary.

10 Claims, 1 Drawing Sheet

Specification includes a Sequence Listing.

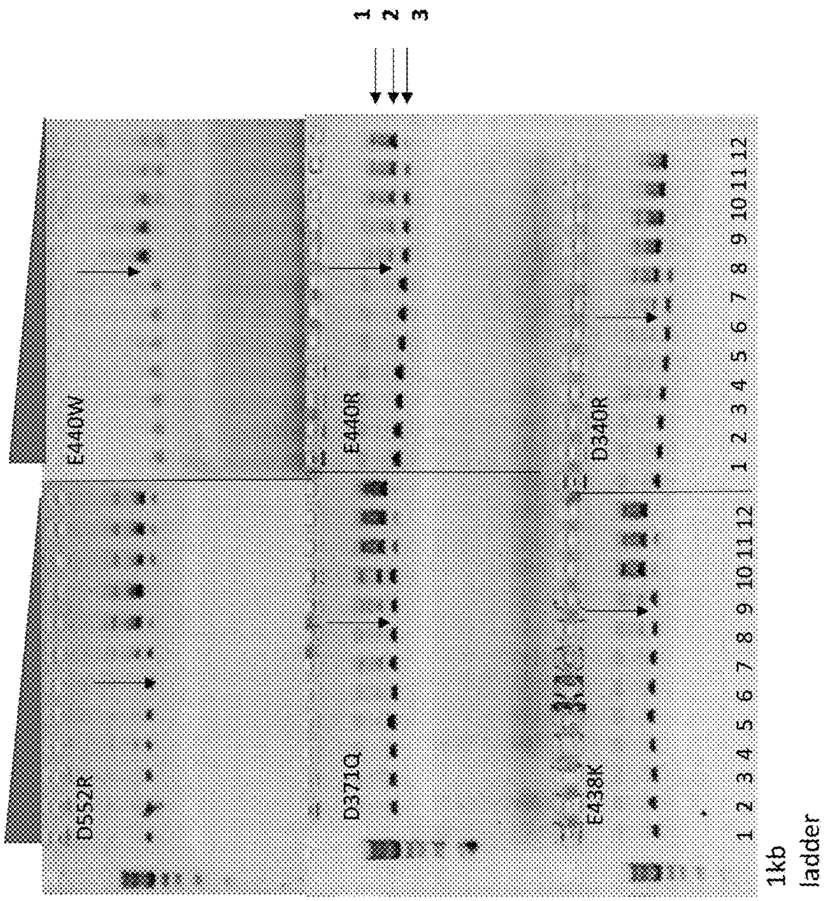
E440W
E440R
D340R
D552R
D371Q
E438K
1kb ladder
1 2 3 4 5 6 7 8 9 10 11 12

Fig 1b

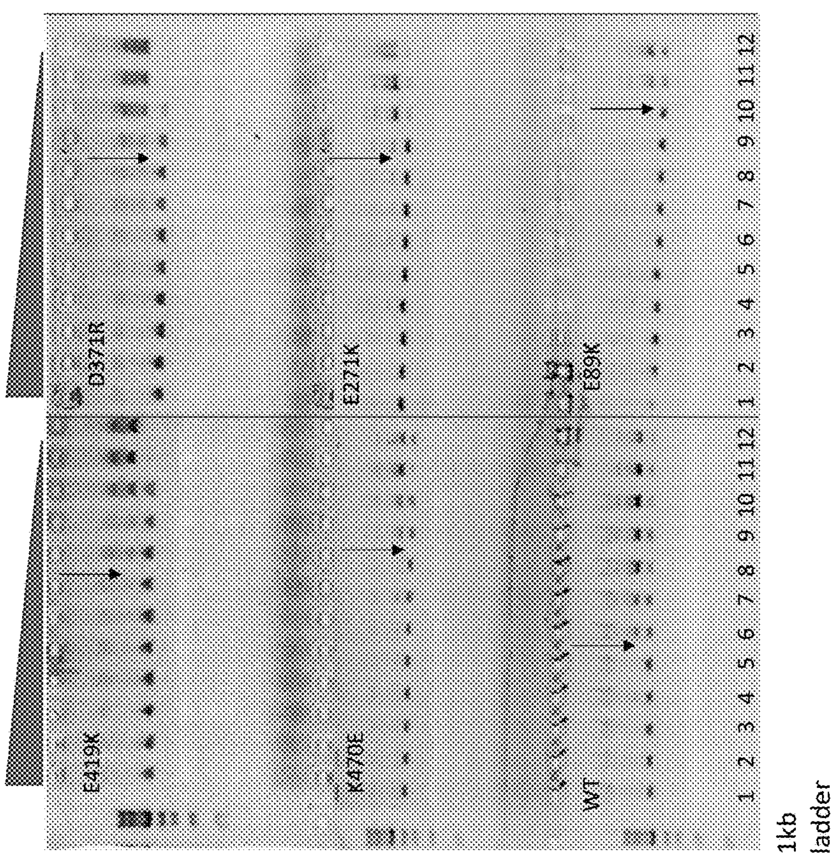
D371R
E271K
E89K
E419K
K470E
WT
1kb ladder
1 2 3 4 5 6 7 8 9 10 11 12

Fig 1a

T4 DNA LIGASE VARIANTS WITH INCREASED LIGATION EFFICIENCY

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Apr. 4, 2022, is named ABCL-T4HiAct_SL.txt and is 78,749 bytes in size.

BACKGROUND

Ligases are commonly used in molecular biology for forming phosphodiester bonds between duplex nucleic acid fragments at the intersection of juxtaposed 5' phosphate and 3' hydroxyl termini. By designing complementary overhangs between each of the double-stranded fragments, ligation can be directed to be both positionally specific and directionally oriented. This allows for specific integration of DNA or RNA fragments into larger vectors to meet the needs of molecular biology research. Out of the commercially available ligases, T4 DNA Ligase is a versatile enzyme that catalyzes the bond joining duplex DNA or RNA at both cohesive ends and blunt ends, has a rapid ligation speed, and also repairs via ligation the mismatches that exist in nicked DNA. It can ligate either cohesive or blunt ends of DNA, RNA and RNA-DNA hybrids (but not single-stranded nucleic acids). It can also ligate blunt-ended DNA with much greater efficiency than *E. coli* DNA ligase.

Because T4 DNA Ligase is the backbone of many molecular biology protocols and is in constant demand, it generates a large share of the revenue for many life science products companies due to its significance in the molecular biology reagent market. Increasing the activity of T4 DNA ligase has benefits for manufacturing of the T4 DNA ligase enzyme itself, allowing fewer production runs to be necessary to create significant final product. Increasing the activity of T4 DNA ligase also increases the usefulness of the enzyme by requiring less enzyme or less total time for a complete ligation of substrates, depending on the goal of the user.

SUMMARY

The invention relates to engineered T4 DNA Ligase mutants exhibiting enhanced ligation activity compared to the wild-type ligase. The following T4 DNA ligase mutants (with a substitution at the position indicated and wherein each mutant's amino acid sequence is the sequence identification number following the indicated substitution, and wherein a DNA sequence for each such mutant is the odd-numbered sequence identification number in the sequence listing preceding the amino acid sequence identification number of each mutant) were identified as having such enhanced ligation activity (where each has a tag sequence: GSGSSGHHHHHH (SEQ ID NO:25) added at the C-Terminus): E89K (SEQ ID NO:4), E271K (SEQ ID NO:6), D340R (SEQ ID NO:8), D371Q (SEQ ID NO:10), D371R (SEQ ID NO:12), E419K (SEQ ID NO:14), E438K (SEQ ID NO:16), E440R (SEQ ID NO:18), E440W (SEQ ID NO:20), D452R (SEQ ID NO:22), and K470E (SEQ ID NO:24) The invention further includes T4 DNA ligase mutant amino acid sequences with at least one of the mutations above, but wherein the remainder of the T4 DNA ligase mutant amino acid sequence only has conservative substitutions such that the molecule has at least 70%, 80%, 90%, 95%, 96%, 97%, 98% or 99% identity to the corresponding T4 DNA ligase mutant amino acid sequence in the sequence listing (hereinafter referred to as "Variant Sequences").

The invention further includes the DNA sequences preceding each of the amino acid sequences for the mutants above (i.e., respectively, SEQ ID NOS:) and further includes the foregoing DNA sequences and other degenerate nucleic acid sequences (collectively the "Degenerate Nucleic Acid Sequences") encoding (i) each of the above T4 DNA ligase mutants, and (ii) the amino acid sequences of any of the Variant Sequences.

The invention further includes vectors incorporating any Degenerate Nucleic Acid Sequences; and cells transformed with any such vectors or Degenerate Nucleic Acid Sequences and capable of expressing any of the above T4 DNA ligase mutant amino acid sequences or Variant Sequences.

The invention further includes a composition or a kit comprising any of the above T4 DNA ligase mutant amino acid sequences or Variant Sequences, Degenerate Nucleic Acid Sequences, or vectors incorporating such Degenerate Nucleic Acid Sequences. The invention also includes a process of amplifying a target nucleic acid, wherein any of the above T4 DNA ligase mutants or Variant Sequences are employed in a reaction mixture designed to amplify a target nucleic acid, and subjecting the reagent mixture to conditions for amplification of the target nucleic acid.

The above mutant T4 DNA ligase mutants have greater activity at lower concentrations in amplifying target DNA sequences compared with wild type, and the Variant Sequences are also expected to have such greater activity.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1**A and **1**B show a series of gel electrophoresis results from an activity assay of the wild type T4 DNA ligase ("WT" in the lower left panel of FIG. **1***a*) compared with each of the T4 DNA ligase mutants. The 1 kb ladder at the left of the three left-most panels in each FIGS. **1***a* and **1***b* is a reference set of gel bands. There are 12 columns in each gel, such that left to right, each column represents the concentration of T4 DNA ligase following a 1:2 serial dilution, and wherein in column 1, 700 ng/μl of T4 DNA Ligase (or mutant) was initially present in the solution. Each gel has an arrow at the dilution level where significant amounts (upper band, labeled "1" on right) of supercoiled plasmid product and restriction digested linearized substrate plasmid (middle band, labeled "2" on right) are apparent. The lowermost band (labeled "3" on right) is the open-circle plasmid product ligated by the T4 DNA ligase wild type or the mutant indicated.

DETAILED DESCRIPTION

The term "biologically active fragment" refers to any fragment, derivative, homolog or analog of a T4 DNA ligase mutant or Variant Sequences that possesses in vivo or in vitro activity that is characteristic of that biomolecule; including, for example, ligase activity or repairing via ligation the mismatches that exist in nicked DNA. In some embodiments, the biologically active fragment, derivative, homolog or analog of the mutant T4 DNA ligase possesses any degree of the biological activity of the mutant T4 DNA ligase in any in vivo or in vitro assay.

In some embodiments, the biologically active fragment can optionally include any number of contiguous amino acid residues of the mutant T4 DNA ligase or Variant Sequences. The invention also includes the polynucleotides encoding any such biologically active fragment and/or Degenerate Nucleic Acid Sequences.

Biologically active fragments can arise from post transcriptional processing or from translation of alternatively spliced RNAs, or alternatively can be created through engineering, bulk synthesis, or other suitable manipulation. Biologically active fragments include fragments expressed in native or endogenous cells as well as those made in expression systems such as, for example, in bacterial, yeast, plant, insect or mammalian cells.

As used herein, the phrase "conservative amino acid substitution" or "conservative mutation" refers to the replacement of one amino acid by another amino acid with a common property. A functional way to define common properties between individual amino acids is to analyze the normalized frequencies of amino acid changes between corresponding proteins of homologous organisms (Schulz (1979) Principles of Protein Structure, Springer-Verlag). According to such analyses, groups of amino acids can be defined where amino acids within a group exchange preferentially with each other, and therefore resemble each other most in their impact on the overall protein structure (Schulz (1979) supra). Examples of amino acid groups defined in this manner can include: a "charged/polar group" including Glu, Asp, Asn, Gln, Lys, Arg, and His; an "aromatic or cyclic group" including Pro, Phe, Tyr, and Trp; and an "aliphatic group" including Gly, Ala, Val, Leu, Ile, Met, Ser, Thr, and Cys. Within each group, subgroups can also be identified. For example, the group of charged/polar amino acids can be sub-divided into sub-groups including: the "positively-charged sub-group" comprising Lys, Arg and His; the "negatively-charged sub-group" comprising Glu and Asp; and the "polar sub-group" comprising Asn and Gln. In another example, the aromatic or cyclic group can be sub-divided into sub-groups including: the "nitrogen ring sub-group" comprising Pro, His, and Trp; and the "phenyl sub-group" comprising Phe and Tyr. In another further example, the aliphatic group can be sub-divided into sub-groups including: the "large aliphatic non-polar sub-group" comprising Val, Leu, and Ile; the "aliphatic slightly-polar sub-group" comprising Met, Ser, Thr, and Cys; and the "small-residue sub-group" comprising Gly and Ala. Examples of conservative mutations include amino acid substitutions of amino acids within the sub-groups above, such as, but not limited to: Lys for Arg or vice versa, such that a positive charge can be maintained; Glu for Asp or vice versa, such that a negative charge can be maintained; Ser for Thr or vice versa, such that a free —OH can be maintained; and Gln for Asn or vice versa, such that a free —NH2 can be maintained. A "conservative variant" is a polypeptide that includes one or more amino acids that have been substituted to replace one or more amino acids of the reference polypeptide (for example, a polypeptide whose sequence is disclosed in a publication or sequence database, or whose sequence has been determined by nucleic acid sequencing) with an amino acid having common properties, e.g., belonging to the same amino acid group or sub-group as delineated above.

When referring to a gene, "mutant" means the gene has at least one base (nucleotide) change, deletion, or insertion with respect to a native or wild type gene. The mutation (change, deletion, and/or insertion of one or more nucleotides) can be in the coding region of the gene or can be in an intron, 3' UTR, 5' UTR, or promoter region. As nonlimiting examples, a mutant gene can be a gene that has an insertion within the promoter region that can either increase or decrease expression of the gene; can be a gene that has a deletion, resulting in production of a nonfunctional protein, truncated protein, dominant negative protein, or no protein; or, can be a gene that has one or more point mutations leading to a change in the amino acid of the encoded protein or results in aberrant splicing of the gene transcript.

The terms "mutant T4 DNA ligase of the invention" and "mutant T4 DNA ligase" when used in this Detailed Description section refer to, depending on the context, collectively or individually, the mutant T4 DNA Ligase polypeptides tested and exhibiting enhanced ligation activity which are:

E89K (SEQ ID NO:4), E271K (SEQ ID NO:6), D340R (SEQ ID NO:8), D371Q (SEQ ID NO:10), D371R (SEQ ID NO:12), E419K (SEQ ID NO:14), E438K (SEQ ID NO:16), E440R (SEQ ID NO:18), E440W (SEQ ID NO:20), D452R (SEQ ID NO:22), and K470E (SEQ ID NO:24), where each has a tag sequence: GSGSSGHHHHHH (SEQ ID NO:25) added at the C-Terminus), and/or Variant Sequences and/or Degenerate Nucleic Acid Sequences, as those terms are defined in the Summary section.

"Naturally-occurring" or "wild-type" refers to the form found in nature. For example, a naturally occurring or wild-type polypeptide or polynucleotide sequence is a sequence present in an organism, which has not been intentionally modified by human manipulation.

The terms "percent identity" or "homology" with respect to nucleic acid or polypeptide sequences are defined as the percentage of nucleotide or amino acid residues in the candidate sequence that are identical with the known polypeptides, after aligning the sequences for maximum percent identity and introducing gaps, if necessary, to achieve the maximum percent homology. N-terminal or C-terminal insertion or deletions shall not be construed as affecting homology. Homology or identity at the nucleotide or amino acid sequence level can be determined by BLAST (Basic Local Alignment Search Tool) analysis using the algorithm employed by the programs blastp, blastn, blastx, tblastn, and tblastx (Altschul (1997), Nucleic Acids Res. 25, 3389-3402, and Karlin (1990), Proc. Natl. Acad. Sci. USA 87, 2264-2268), which are tailored for sequence similarity searching. The approach used by the BLAST program is to first consider similar segments, with and without gaps, between a query sequence and a database sequence, then to evaluate the statistical significance of all matches that are identified, and finally to summarize only those matches which satisfy a preselected threshold of significance. For a discussion of basic issues in similarity searching of sequence databases, see Altschul (1994), Nature Genetics 6, 119-129. The search parameters for histogram, descriptions, alignments, expect (i.e., the statistical significance threshold for reporting matches against database sequences), cutoff, matrix, and filter (low complexity) can be at the default settings. The default scoring matrix used by blastp, blastx, tblastn, and tblastx is the BLOSUM62 matrix (Henikoff (1992), Proc. Natl. Acad. Sci. USA 89, 10915-10919), recommended for query sequences over 85 units in length (nucleotide bases or amino acids).

In some embodiments, the invention relates to methods (and related kits, systems, apparatuses and compositions) for performing a ligation reaction comprising or consisting of contacting a mutant T4 DNA ligase or a biologically active fragment thereof with a nucleic acid template in the presence of one or more nucleotides, and ligating at least one of the one or more nucleotides using the mutant T4 DNA ligase or the biologically active fragment thereof.

In some embodiments, the method can include ligating a double stranded RNA or DNA polynucleotide strand into a circular molecule. In some embodiments, the method can further include detecting a signal indicating the ligation by using a sensor. In some embodiments, the sensor is an ISFET. In some embodiments, the sensor can include a detectable label or detectable reagent within the ligating reaction.

In some embodiments, the invention relates to methods (and related kits, systems, apparatus and compositions) for performing rolling circle amplification (see U.S. Pat. No. 5,714,320, incorporated by reference) of a nucleic acid, using the mutant T4 DNA ligase as the enzyme in the ligation steps of the amplification process. The amplifying includes amplifying the nucleic acid in solution, as well as clonally amplifying the nucleic acid on a solid support such as a nucleic acid bead, flow cell, nucleic acid array, or wells present on the surface of the solid support.

Making Mutant T4 DNA Ligase

The mutant T4 DNA ligase of the invention can be expressed in any suitable host system, including a bacterial, yeast, fungal, baculovirus, plant or mammalian host cell. For bacterial host cells, suitable promoters for directing transcription of the nucleic acid constructs of the present disclosure, include the promoters obtained from the *E. coli* lac operon, *Streptomyces coelicolor* agarase gene (dagA), *Bacillus subtilis* levansucrase gene (sacB), *Bacillus licheniformis* alpha-amylase gene (amyL), *Bacillus stearothermophilus* maltogenic amylase gene (amyM), *Bacillus amyloliquefaciens* alpha-amylase gene (amyQ), *Bacillus licheniformis* penicillinase gene (penP), *Bacillus subtilis* xylA and xylB genes, and prokaryotic beta-lactamase gene (Villa-Kamaroff et al., 1978, Proc. Natl Acad. Sci. USA 75: 3727-3731), as well as the tac promoter (DeBoer et al., 1983, Proc. Natl Acad. Sci. USA 80: 21-25).

For filamentous fungal host cells, suitable promoters for directing the transcription of the nucleic acid constructs of the present disclosure include promoters obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Rhizomucor miehei* aspartic proteinase, *Aspergillus niger* neutral alpha-amylase, *Aspergillus niger* acid stable alpha-amylase, *Aspergillus niger* or *Aspergillus awamori* glucoamylase (glaA), *Rhizomucor miehei* lipase, *Aspergillus oryzae* alkaline protease, *Aspergillus oryzae* triose phosphate isomerase, *Aspergillus nidulans* acetamidase, and *Fusarium oxysporum* trypsin-like protease (WO 96/00787), as well as the NA2-tpi promoter (a hybrid of the promoters from the genes for *Aspergillus niger* neutral alpha-amylase and *Aspergillus oryzae* triose phosphate isomerase), and mutant, truncated, and hybrid promoters thereof.

In a yeast host, useful promoters can be from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* galactokinase (GAL1), *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP), and *Saccharomyces cerevisiae* 3-phosphoglycerate kinase. Other useful promoters for yeast host cells are described by Romanos et al., 1992, Yeast 8:423-488.

For baculovirus expression, insect cell lines derived from Lepidopterans (moths and butterflies), such as *Spodoptera frugiperda*, are used as host. Gene expression is under the control of a strong promoter, e.g., pPolh.

Plant expression vectors are based on the Ti plasmid of *Agrobacterium tumefaciens*, or on the tobacco mosaic virus (TMV), potato virus X, or the cowpea mosaic virus. A commonly used constitutive promoter in plant expression vectors is the cauliflower mosaic virus (CaMV) 35S promoter.

For mammalian expression, cultured mammalian cell lines such as the Chinese hamster ovary (CHO), COS, including human cell lines such as HEK and HeLa may be used to produce the mutant T4 DNA ligase. Examples of mammalian expression vectors include the adenoviral vectors, the pSV and the pCMV series of plasmid vectors, vaccinia and retroviral vectors, as well as baculovirus. The promoters for cytomegalovirus (CMV) and SV40 are commonly used in mammalian expression vectors to drive gene expression. Non-viral promoters, such as the elongation factor (EF)-1 promoter, are also known.

The control sequence for the expression may also be a suitable transcription terminator sequence, that is, a sequence recognized by a host cell to terminate transcription. The terminator sequence is operably linked to the 3' terminus of the nucleic acid sequence encoding the polypeptide. Any terminator which is functional in the host cell of choice may be used.

For example, exemplary transcription terminators for filamentous fungal host cells can be obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* glucoamylase, *Aspergillus nidulans* anthranilate synthase, *Aspergillus niger* alpha-glucosidase, and *Fusarium oxysporum* trypsin-like protease.

Exemplary terminators for yeast host cells can be obtained from the genes for *Saccharomyces cerevisiae* enolase, *Saccharomyces cerevisiae* cytochrome C (CYC1), and *Saccharomyces cerevisiae* glyceraldehyde-3-phosphate dehydrogenase.

Terminators for insect, plant and mammalian host cells are also well known.

The control sequence may also be a suitable leader sequence, a nontranslated region of an mRNA that is important for translation by the host cell. The leader sequence is operably linked to the 5' terminus of the nucleic acid sequence encoding the polypeptide. Any leader sequence that is functional in the host cell of choice may be used. Exemplary leaders for filamentous fungal host cells are obtained from the genes for *Aspergillus oryzae* TAKA amylase and *Aspergillus nidulans* triose phosphate isomerase. Suitable leaders for yeast host cells are obtained from the genes for *Saccharomyces cerevisiae* enolase (ENO-1), *Saccharomyces cerevisiae* 3-phosphoglycerate kinase, *Saccharomyces cerevisiae* alpha-factor, and *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP).

The control sequence may also be a polyadenylation sequence, a sequence operably linked to the 3' terminus of the nucleic acid sequence and which, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA. Any polyadenylation sequence which is functional in the host cell of choice may be used in the present invention. Exemplary polyadenylation sequences for filamentous fungal host cells can be from the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* glucoamylase, *Aspergillus nidulans* anthranilate synthase, *Fusarium oxysporum* tryp sin-like protease, and *Aspergillus niger* alpha-glucosidase.

The control sequence may also be a signal peptide coding region that codes for an amino acid sequence linked to the amino terminus of a polypeptide and directs the encoded polypeptide into the cell's secretory pathway. The 5' end of the coding sequence of the nucleic acid sequence may inherently contain a signal peptide coding region naturally linked in translation reading frame with the segment of the coding region that encodes the secreted polypeptide. Alternatively, the 5' end of the coding sequence may contain a signal peptide coding region that is foreign to the coding sequence. The foreign signal peptide coding region may be required where the coding sequence does not naturally contain a signal peptide coding region.

Alternatively, the foreign signal peptide coding region may simply replace the natural signal peptide coding region in order to enhance secretion of the polypeptide. However, any signal peptide coding region which directs the expressed polypeptide into the secretory pathway of a host cell of choice may be used.

Effective signal peptide coding regions for bacterial host cells are the signal peptide coding regions obtained from the genes for *Bacillus* NCIB 11837 maltogenic amylase, *Bacillus stearothermophilus* alpha-amylase, *Bacillus licheniformis* subtilisin, *Bacillus licheniformis* beta-lactamase, *Bacillus stearothermophilus* neutral proteases (nprT, nprS, nprM), and *Bacillus subtilis* prsA. Further signal peptides are described by Simonen and Palva, 1993, Microbiol Rev 57: 109-137.

Effective signal peptide coding regions for filamentous fungal host cells can be the signal peptide coding regions obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Aspergillus niger* neutral amylase, *Aspergillus niger* glucoamylase, *Rhizomucor miehei* aspartic proteinase, *Humicola insolens* cellulase, and *Humicola lanuginosa* lipase.

Useful signal peptides for yeast host cells can be from the genes for *Saccharomyces cerevisiae* alpha-factor and *Saccharomyces cerevisiae* invertase. Signal peptides for other host cell systems are also well known.

The control sequence may also be a propeptide coding region that codes for an amino acid sequence positioned at the amino terminus of a polypeptide. The resultant polypeptide is known as a proenzyme or propolypeptide (or a zymogen in some cases). A propolypeptide is generally inactive and can be converted to a mature active polypeptide by catalytic or autocatalytic cleavage of the propeptide from the propolypeptide. The propeptide coding region may be obtained from the genes for *Bacillus subtilis* alkaline protease (aprE), *Bacillus subtilis* neutral protease (nprT), *Saccharomyces cerevisiae* alpha-factor, *Rhizomucor miehei* aspartic proteinase, and *Myceliophthora thermophila* lactase (WO 95/33836).

Where both signal peptide and propeptide regions are present at the amino terminus of a polypeptide, the propeptide region is positioned next to the amino terminus of a polypeptide and the signal peptide region is positioned next to the amino terminus of the propeptide region.

It may also be desirable to add regulatory sequences, which allow the regulation of the expression of the mutant T4 DNA ligase relative to the growth of the host cell. Examples of regulatory systems are those which cause the expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound. In prokaryotic host cells, suitable regulatory sequences include the lac, tac, and trp operator systems. In yeast host cells, suitable regulatory systems include, as examples, the ADH2 system or GAL1 system. In filamentous fungi, suitable regulatory sequences include the TAKA alpha-amylase promoter, *Aspergillus niger* glucoamylase promoter, and *Aspergillus oryzae* glucoamylase promoter. Regulatory systems for other host cells are also well known.

Other examples of regulatory sequences are those which allow for gene amplification. In eukaryotic systems, these include the dihydrofolate reductase gene, which is amplified in the presence of methotrexate, and the metallothionein genes, which are amplified with heavy metals. In these cases, the nucleic acid sequence encoding the KRED polypeptide of the present invention would be operably linked with the regulatory sequence.

Another embodiment includes a recombinant expression vector comprising a polynucleotide encoding an engineered mutant T4 DNA ligase or a variant thereof, and one or more expression regulating regions such as a promoter and a terminator, and a replication origin, depending on the type of hosts into which they are to be introduced. The various nucleic acid and control sequences described above may be joined together to produce a recombinant expression vector which may include one or more convenient restriction sites to allow for insertion or substitution of the nucleic acid sequence encoding the mutant T4 DNA ligase at such sites. Alternatively, the nucleic acid sequences of the mutant T4 DNA ligase may be expressed by inserting the nucleic acid sequences or a nucleic acid construct comprising the sequences into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

The recombinant expression vector may be any vector (e.g., a plasmid or virus), which can be conveniently subjected to recombinant DNA procedures and can bring about the expression of the mutant T4 DNA ligase polynucleotide sequence. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vectors may be linear or closed circular plasmids.

The expression vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one which, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids which together contain the total DNA to be introduced into the genome of the host cell, or a transposon may be used.

The expression vector herein preferably contain one or more selectable markers, which permit easy selection of transformed cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like. Examples of bacterial selectable markers are the dal genes from *Bacillus subtilis* or *Bacillus licheniformis*, or markers, which confer antibiotic resistance such as ampicillin, kanamycin, chloramphenicol (Example 1) or tetracycline resistance. Suitable markers for yeast host cells are ADE2, HIS3, LEU2, LYS2, METS, TRP1, and URA3. Selectable markers for use in a filamentous fungal host cell include, but are not limited to, amdS (acetamidase), argB (ornithine carbamoyltransferase), bar (phosphinothricin acetyltransferase), hph (hygromycin phosphotransferase), niaD (nitrate reductase), pyrG (orotidine-5'-phosphate decarboxylase), sC (sulfate adenyltransferase), and trpC (anthranilate synthase), as well as equivalents thereof. Embodiments for use in an *Aspergillus* cell include the amdS and pyrG genes of *Asper-*

gillus nidulans or *Aspergillus oryzae* and the bar gene of *Streptomyces hygroscopicus*. Selectable markers for insect, plant and mammalian cells are also well known.

The expression vectors of the present invention preferably contain an element(s) that permits integration of the vector into the host cell's genome or autonomous replication of the vector in the cell independent of the genome. For integration into the host cell genome, the vector may rely on the nucleic acid sequence encoding the polypeptide or any other element of the vector for integration of the vector into the genome by homologous or nonhomologous recombination.

Alternatively, the expression vector may contain additional nucleic acid sequences for directing integration by homologous recombination into the genome of the host cell. The additional nucleic acid sequences enable the vector to be integrated into the host cell genome at a precise location(s) in the chromosome(s). The integrational elements may be any sequence that is homologous with the target sequence in the genome of the host cell. Furthermore, the integrational elements may be non-encoding or encoding nucleic acid sequences. On the other hand, the vector may be integrated into the genome of the host cell by non-homologous recombination.

For autonomous replication, the vector may further comprise an origin of replication enabling the vector to replicate autonomously in the host cell in question. Examples of bacterial origins of replication are P15A ori, or the origins of replication of plasmids pBR322, pUC19, pACYC177 (which plasmid has the P15A ori), or pACYC184 permitting replication in *E. coli*, and pUB 110, pE194, pTA1060, or pAM31 permitting replication in *Bacillus*. Examples of origins of replication for use in a yeast host cell are the 2 micron origin of replication, ARS1, ARS4, the combination of ARS1 and CEN3, and the combination of ARS4 and CEN6. The origin of replication may be one having a mutation which makes it's functioning temperature-sensitive in the host cell (see, e.g., Ehrlich, 1978, Proc Natl Acad Sci. USA 75:1433).

More than one copy of a nucleic acid sequence of the mutant T4 DNA ligase may be inserted into the host cell to increase production of the gene product. An increase in the copy number of the nucleic acid sequence can be obtained by integrating at least one additional copy of the sequence into the host cell genome or by including an amplifiable selectable marker gene with the nucleic acid sequence where cells containing amplified copies of the selectable marker gene, and thereby additional copies of the nucleic acid sequence, can be selected for by cultivating the cells in the presence of the appropriate selectable agent.

Expression vectors for the mutant T4 DNA ligase polynucleotide are commercially available. Suitable commercial expression vectors include p3×FLAG™ expression vectors from Sigma-Aldrich Chemicals, St. Louis Mo., which includes a CMV promoter and hGH polyadenylation site for expression in mammalian host cells and a pBR322 origin of replication and ampicillin resistance markers for amplification in *E. coli*. Other suitable expression vectors are pBluescriptII SK(−) and pBK-CMV, which are commercially available from Stratagene, LaJolla Calif., and plasmids which are derived from pBR322 (Gibco BRL), pUC (Gibco BRL), pREP4, pCEP4 (Invitrogen) or pPoly (Lathe et al., 1987, Gene 57:193-201).

Suitable host cells for expression of a polynucleotide encoding the mutant T4 DNA ligase, are well known in the art and include but are not limited to, bacterial cells, such as *E. coli, Lactobacillus kefir, Lactobacillus brevis, Lactobacillus minor, Streptomyces* and *Salmonella typhimurium* cells; fungal cells, such as yeast cells (e.g., *Saccharomyces cerevisiae* or *Pichia pastoris* (ATCC Accession No. 201178)); insect cells such as *Drosophila* S2 and *Spodoptera* Sf9 cells; animal cells such as CHO, COS, BHK, 293, and Bowes melanoma cells; and plant cells. Appropriate culture mediums and growth conditions for the above-described host cells are well known in the art.

Polynucleotides for expression of the mutant T4 DNA ligase may be introduced into cells by various methods known in the art. Techniques include among others, electroporation, biolistic particle bombardment, liposome mediated transfection, calcium chloride transfection, and protoplast fusion. Various methods for introducing polynucleotides into cells are known to the skilled artisan.

Polynucleotides encoding the mutant T4 DNA ligase can be prepared by standard solid-phase methods, according to known synthetic methods. In some embodiments, fragments of up to about 100 bases can be individually synthesized, then joined (e.g., by enzymatic or chemical litigation methods, or polymerase mediated methods) to form any desired continuous sequence. For example, polynucleotides can be prepared by chemical synthesis using, e.g., the classical phosphoramidite method described by Beaucage et al., 1981, Tet Lett 22:1859-69, or the method described by Matthes et al., 1984, EMBO J. 3:801-05, e.g., as it is typically practiced in automated synthetic methods. According to the phosphoramidite method, oligonucleotides are synthesized, e.g., in an automatic DNA synthesizer, purified, annealed, ligated and cloned in appropriate vectors. In addition, essentially any nucleic acid can be obtained from any of a variety of commercial sources, such as The Midland Certified Reagent Company, Midland, Tex., The Great American Gene Company, Ramona, Calif., ExpressGen Inc. Chicago, Ill., and Operon Technologies Inc., Alameda, Calif.

Engineered the mutant T4 DNA ligase expressed in a host cell can be recovered from the cells and or the culture medium using any one or more of the well-known techniques for protein purification, including, among others, lysozyme treatment, sonication, filtration, salting-out, ultracentrifugation, and chromatography. Suitable solutions for lysing and the high efficiency extraction of proteins from bacteria, such as *E. coli*, are commercially available under the trade name CelLytic B™ from Sigma-Aldrich of St. Louis Mo.

Chromatographic techniques for isolation of the mutant T4 DNA ligase include, among others, reverse phase chromatography high performance liquid chromatography, ion exchange chromatography, gel electrophoresis, and affinity chromatography. Conditions for purification will depend, in part, on factors such as net charge, hydrophobicity, hydrophilicity, molecular weight, molecular shape, and will be apparent to those having skill in the art.

In some embodiments, affinity techniques may be used to isolate the mutant T4 DNA ligase. For affinity chromatography purification, any antibody which specifically binds the mutant T4 DNA ligase may be used. For the production of antibodies, various host animals, including but not limited to rabbits, mice, rats, etc., may be immunized by injection with a compound. The compound may be attached to a suitable carrier, such as BSA, by means of a side chain functional group or linkers attached to a side chain functional group. Various adjuvants may be used to increase the immunological response, depending on the host species, including but not limited to Freund's (complete and incomplete), mineral gels such as aluminum hydroxide, surface active substances such as lysolecithin, pluronic polyols, polyanions, peptides, oil emulsions, keyhole limpet hemocyanin, dinitrophenol, and potentially useful human adjuvants such as BCG (bacilli Calmette Guerin) and *Corynebacterium parvum*.

Example of Making T4 DNA Ligase Mutants

T4 DNA ligase mutants were generated by conventional PCR mutagenesis wherein the primers are designed to include the desired base substitution, and during PCR, the mutation is incorporated into the amplicon, replacing the original sequence. All T4 DNA ligase mutants and the wild type had an added C-terminal 6-mer His tag (SEQ ID NO:26) for ease of purification, preceded by a 6-mer series of Ser and Gly residues as shown.

PCR was followed with a DpnI digestion which destroys the methylated template (that doesn't contain the substitution) leaving behind only the unmethylated PCR amplicons with the substitution.

The PCR amplicons are then then directly transformed into chemically competent *E. Coli* host cells, wherein the bacteria were pre-treated with chemicals to enable them to take up plasmids incorporating the amplicons. See ThermoFisher Scientific, Chemically Competent Cells webpage (featuring kits for generating chemically competent cells).

The mutant T4 DNA ligase polypeptides expressed from the transformed *E. Coli* host cells were characterized and selected based on a standard ligation assay with gel electrophoresis. Ligase catalyzes the formation of phosphodiester bonds between the 5' and 3' ends of complementary cohesive ends or blunt ends of duplex DNA, and the extent of ligation with different T4 DNA ligase mutants can be visualized on an agarose gel using appropriate DNA dyes. In this case, GelRed (Biotium, San Francisco CA) was used to stain the gels for visualization under UV light. Examining each T4 DNA ligase mutant's performance based on their ligation activity at diminishing concentrations of enzyme, and comparing the resultant activity to that of similarly diluted wild type ("WT") ligase, allowed determination of which mutants showed increased activity compared to the wild-type under the same conditions.

The ligation substrate for characterization of the mutant T4 DNA ligase polypeptides expressed from the transformed *E. Coli* host cells was prepared as follows.

The DNA vector used was pUC19 (New England Bio Labs, catalog number N3041S). PUC19 is double stranded circle that is 2686 base pairs long. PUC19 was digested with BsaI-HF®v2 (New England Bio Labs, catalog number R3733S) which has an offset/nonsymmetrical cleavage site, where the 5' strand is cleaved after the recognition sequence plus the addition of one random (N1) nucleotide, and on the 3' strand cleavage occurs after the complementary recognition sequence plus five additional random (N5) nucleotides. The 5' recognition sequence is GGTCTC. The cleavage site of PUC19 by BsaI-HF®v2 (which is a high fidelity version of the restriction enzyme BsaI) is designated 5'-GGTCTC (N1)/(N5)-3'.

5 μl of puC19 at a concentration of 1 mg/ml was combined with 2.5 μl of 20,000 units/ml BsaI-HF®v2 v2 (which is a high fidelity version of the restriction enzyme BsaI), 5 μl 10×rCutSmart™ Buffer (New England Bio Labs, catalog number B6004S) (50 mM Potassium Acetate, 20 mM Tris-acetate, 10 mM Magnesium Acetate, 100 μg/ml Recombinant Albumin), and 35 μl of water. After all components were combined, the composition was incubated at 37° C. for the digestion to take place. After one hour at 37° C., the reaction was incubated at 80° C. for 20 minutes to heat inactivate the BsaI-HF®v2 (which is a high fidelity version of the restriction enzyme BsaI). The mixture was then diluted with water to a concentration of 10 ng/μl.

Example: Ligation Assay and Results

The ligation procedure was performed as follows. Each T4 DNA ligase, either wild-type or variant, was diluted with enzyme diluent (50% glycerol, 10 mM tris-HCl) in a serial dilution such that for each sample 12 different concentrations were obtained. The starting concentration for each serial dilution was 700 ng/μ, diluted two fold in each next dilution such that the final concentration was 50% as concentrated as the sample before it, and so on for a total of 12 1:2 dilutions. Four μl of each enzyme or serial dilution was then added to a PCR plate.

To the enzyme, 16 μl master mix consisting of 2 μl of 10×T4 DNA Ligase Reaction Buffer (New England Biolabs, catalog number B0202A; including 50 mM Tris-HCl, 10 mM $MgCl_2$, 1 mM ATP, 10 mM DTT (dithiothreitol)), 1 μl of 10 ng/μl pUC19 digested with BsaI-HF®v2 (which is a high fidelity version of the restriction enzyme BsaI), and 13 μl of water was added to each reaction. This makes the total volume per reaction up to 20 μl. The reactions were incubated at 16° C. for 10 minutes. After the 10-minute incubation period the reactions received a heat shock of 80° C. for 2 minutes to stop any further activity. 6 μl of stop solution (120 mM EDTA, 30% glycerol, 50 mM Tris-HCl pH 8.0, 0.0125% bromophenol blue, 0.1% SDS, and 5× Gel Red Nucleic Acid Stain (Biotium, Fremont, CA)) was added to each reaction.

Gel electrophoresis with 1.2% agarose gel was used to visualize the ligation reaction product. Each gel had a wild-type T4 DNA ligase sample series and 7 variant T4 DNA ligase sample series. Each gel was run for 35 minutes at 180V.

Results as compared with wild type are shown in FIGS. **1**A and **1**B. The 12 mutants identified that exhibit increased ligation activity are the following:

E89K (SEQ ID NO:4), E271K (SEQ ID NO:6), D340R (SEQ ID NO:8), D371Q (SEQ ID NO:10), D371R (SEQ ID NO:12), E419K (SEQ ID NO:14), E438K (SEQ ID NO:16), E440R (SEQ ID NO:18), E440W (SEQ ID NO:20), D452R (SEQ ID NO:22), and K470E (SEQ ID NO:24), where each has a tag sequence: GSGSSGHHHHHH (SEQ ID NO:25) added at the C-Terminus, as shown in FIGS. **1**A, **1**B.

Comparative activity results are tabulated in Table 1 below, showing lane differences for each T4 DNA ligase mutant, where each lane difference greater than WT is assigned a 2-fold activity increase. For example, 1 lane activity improvement over WT was given the value 2 (as shown by little or no apparent upper band representing supercoiled plasmid product and little or no apparent middle band representing restriction digested linearized substrate plasmid in a lane), while 2 lanes activity improvement over WT was given the value 4 and so on.

TABLE 1

T4 DNA Ligase variants with greater activity than WT

Mutants	Activity vs. WT (Fold)
E89K	32
K271K	8
D340R	2
D371R	8
D371Q	4
E419K	4

TABLE 1-continued

T4 DNA Ligase variants with greater activity than WT

Mutants	Activity vs. WT (Fold)
E438K	8
E440R	4
E440W	4
K470E	8
D452R	2

SEQ ID NO: 1 T4 DNA Ligase CH Wild Type DNA

1	ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA	60
61	ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT	120
121	CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT	180
181	GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA	240
241	TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT	300
301	GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT	360
361	ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT	420
421	TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT	480
481	GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA	540
541	TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG	600
601	ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC	660
661	CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA	720
721	AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC	780
781	GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG	840
841	GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT	900
901	GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT	960
961	GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC	1020
1021	CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA	1080
1081	AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT	1140
1141	CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA	1200
1201	ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT	1260
1261	GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA	1320
1321	TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT	1380
1381	GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT	1440
1441	TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA	1500

SEQ ID NO: 2 T4 DNA Ligase CH Wild Type Protein

1	MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF	60
61	GMLTLTDMLDFIEFTLATRKLIGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS	120
121	IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL	180
181	SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE	240
241	NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY	300
301	DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS	360
361	KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL	420

-continued

421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

SEQ ID NO: 3 T4 DNA Ligase CH E89K DNA

1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA 60

61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT 120

121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT 180

181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA 240

241 TTGACTGGAAATGCAGCAATTGAAAAATTAACTGGATATATCACCGATGGTAAAAAAGAT 300

301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT 360

361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG 600

601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC 660

661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA 720

721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC 780

781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT 1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA 1500

SEQ ID NO: 4 T4 DNA Ligase CH E89K Protein

1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF 60

61 GMLTLTDMLDFIEFTLATRKLIGNAAIEKLTGYITDGKKDDVEVLRRVMMRDLECGASVS 120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL 180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE 240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY 300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS 360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL 420

421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

-continued

SEQ ID NO: 5 T4 DNA Ligase CH E271K DNA

1	ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA	60
61	ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT	120
121	CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT	180
181	GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA	240
241	TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT	300
301	GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT	360
361	ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT	420
421	TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT	480
481	GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA	540
541	TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG	600
601	ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC	660
661	CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA	720
721	AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC	780
781	GCCAATAAATCTTTAAAGGGAACCATTTCTAAAAAAGAAGCACAATGCATGAAGTTTCAG	840
841	GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT	900
901	GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT	960
961	GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC	1020
1021	CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA	1080
1081	AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT	1140
1141	CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA	1200
1201	ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT	1260
1261	GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA	1320
1321	TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT	1380
1381	GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT	1440
1441	TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA	1500

SEQ ID NO: 6 T4 DNA Ligase CH E271K Protein

1	MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF	60
61	GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS	120
121	IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL	180
181	SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE	240
241	NSKAKEFAEVAESRTASNGIANKSLKGTISKKEAQCMKFQVWDYVPLVEIYSLPAFRLKY	300
301	DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS	360
361	KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL	420
421	DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD	480
481	FHEVTGLGSGSSGHHHHHH*	499

SEQ ID NO: 7 T4 DNA Ligase CH D340R DNA

1	ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA	60
61	ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT	120

-continued

121	CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT	180
181	GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA	240
241	TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT	300
301	GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT	360
361	ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT	420
421	TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT	480
481	GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA	540
541	TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG	600
601	ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC	660
661	CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA	720
721	AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC	780
781	GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG	840
841	GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT	900
901	GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT	960
961	GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTCGT	1020
1021	CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA	1080
1081	AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT	1140
1141	CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA	1200
1201	ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT	1260
1261	GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA	1320
1321	TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT	1380
1381	GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT	1440
1441	TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA	1500

SEQ ID NO: 8 T4 DNA Ligase CH D340R Protein

1	MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF	60
61	GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS	120
121	IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL	180
181	SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE	240
241	NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY	300
301	DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIRQGLEGIILKNIDGLWENARS	360
361	KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL	420
421	DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD	480
481	FHEVTGLGSGSSGHHHHHH*	499

SEQ ID NO: 9 T4 DNA Ligase CH D371Q DNA

1	ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA	60
61	ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT	120
121	CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT	180
181	GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA	240
241	TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT	300
301	GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT	360

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361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG 600

601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC 660

661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA 720

721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC 780

781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTCAATTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT 1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA 1500

SEQ ID NO: 10 T4 DNA Ligase CH D371Q Protein

1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF 60

61 GMLTLTDMLDFIEFTLATRKLIGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS 120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL 180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE 240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY 300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS 360

361 KNLYKFKEVIQVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL 420

421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

SEQ ID NO: 11 T4 DNA Ligase CH D371R DNA

1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA 60

61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT 120

121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT 180

181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA 240

241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT 300

301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT 360

361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

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 541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG   600
 601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC   660
 661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA   720
 721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC   780
 781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG   840
 841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT   900
 901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT   960
 961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC  1020
1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA  1080
1081 AAAAATCTTTATAAATTTAAAGAAGTAATTCGTGTTGATTTAAAAATTGTAGGAATTTAT  1140
1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA  1200
1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT  1260
1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA  1320
1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT  1380
1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT  1440
1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA  1500
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SEQ ID NO: 12 T4 DNA Ligase CH D371R Protein

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   1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF     60
  61 GMLTLTDMLDFIEFTLATRKLIGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS    120
 121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL    180
 181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE    240
 241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY    300
 301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS    360
 361 KNLYKFKEVIRVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL    420
 421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD    480
 481 FHEVTGLGSGSSGHHHHHH*                                            499
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SEQ ID NO: 13 T4 DNA Ligase CH E419K DNA

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   1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA     60
  61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT    120
 121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT    180
 181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA    240
 241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT    300
 301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT    360
 361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT    420
 421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT    480
 481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA    540
 541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG    600
 601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC    660
 661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA    720
 721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC    780
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781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATAAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT 1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA 1500

SEQ ID NO: 14 T4 DNA Ligase CH E419K Protein

1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF 60

61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS 120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL 180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE 240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY 300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS 360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHKL 420

421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

SEQ ID NO: 15 T4 DNA Ligase CH E438K DNA

1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA 60

61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT 120

121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT 180

181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA 240

241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT 300

301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT 360

361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG 600

601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC 660

661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA 720

721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC 780

781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

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961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAAAATGCGAA 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT 1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA 1500

SEQ ID NO: 16 T4 DNA Ligase CH E438K Protein

1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF 60

61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS 120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL 180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE 240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY 300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS 360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL 420

421 DRTRIMENQNYYIGKILKCECNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

SEQ ID NO: 17 T4 DNA Ligase CH E440R DNA

1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA 60

61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT 120

121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT 180

181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA 240

241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT 300

301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT 360

361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG 600

601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC 660

661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA 720

721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC 780

781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

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1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCCGT 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT 1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA 1500

SEQ ID NO: 18 T4 DNA Ligase CH E440R Protein

1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF 60

61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS 120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL 180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE 240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY 300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS 360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL 420

421 DRTRIMENQNYYIGKILECRCNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD 480

481 FHEVTGLGSGSSGHHHHHH* 499

SEQ ID NO: 19 T4 DNA Ligase CH E440W DNA

1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA 60

61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT 120

121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT 180

181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA 240

241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT 300

301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT 360

361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT 420

421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT 480

481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA 540

541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG 600

601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC 660

661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA 720

721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC 780

781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG 840

841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT 900

901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT 960

961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC 1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA 1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT 1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA 1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT 1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCTGG 1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT 1380

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1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT   1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA   1500
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SEQ ID NO: 20 T4 DNA Ligase CH E440W Protein

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  1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF    60

 61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS   120

121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL   180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE   240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY   300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS   360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL   420

421 DRTRIMENQNYYIGKILECWCNGWLKSDGRTDYVKLFLPIAIRLREDKTKANTFEDVFGD   480

481 FHEVTGLGSGSSGHHHHHH*                                           499
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SEQ ID NO: 21 T4 DNA Ligase CH D452R DNA

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   1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA     60

  61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT    120

 121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT    180

 181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA    240

 241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT    300

 301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT    360

 361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT    420

 421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT    480

 481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA    540

 541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG    600

 601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC    660

 661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA    720

 721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC    780

 781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG    840

 841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT    900

 901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT    960

 961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC   1020

1021 CAAGGTCTTGAAGGTATTATTCTCAAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA  1080

1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT   1140

1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA   1200

1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT   1260

1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA   1320

1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTCGTTACGTTAAATTATTTCTTCCGATT   1380

1381 GCGATTCGTTTACGTGAAGATAAAACTAAAGCTAATACATTCGAAGATGTATTTGGTGAT   1440

1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA   1500
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SEQ ID NO: 22 T4 DNA Ligase CH D452R Protein

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  1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF    60
 61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS   120
121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL   180
181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE   240
241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY   300
301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS   360
361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL   420
421 DRTRIMENQNYYIGKILECECNGWLKSDGRTRYVKLFLPIAIRLREDKTKANTFEDVFGD   480
481 FHEVTGLGSGSSGHHHHHH*                                           499
```

SEQ ID NO: 23 T4 DNA Ligase CH K470E DNA

```
   1 ATGATTCTTAAAATTCTGAACGAAATAGCATCTATTGGTTCAACTAAACAGAAGCAAGCA    60
  61 ATTCTTGAAAAGAATAAAGATAATGAATTGCTTAAACGAGTATATCGTCTGACTTATTCT   120
 121 CGTGGGTTACAGTATTATATCAAGAAATGGCCTAAACCTGGTATTGCTACCCAGAGTTTT   180
 181 GGAATGTTGACTCTTACCGATATGCTTGACTTCATTGAATTCACATTAGCTACTCGGAAA   240
 241 TTGACTGGAAATGCAGCAATTGAGGAATTAACTGGATATATCACCGATGGTAAAAAAGAT   300
 301 GATGTTGAAGTTTTGCGTCGAGTGATGATGCGAGACCTTGAATGTGGTGCTTCAGTATCT   360
 361 ATTGCAAACAAAGTTTGGCCAGGTTTAATTCCTGAACAACCTCAAATGCTCGCAAGTTCT   420
 421 TATGATGAAAAAGGCATTAATAAGAATATCAAATTTCCAGCCTTTGCTCAGTTAAAAGCT   480
 481 GATGGAGCTCGGTGTTTTGCTGAAGTTAGAGGTGATGAATTAGATGATGTTCGTCTTTTA   540
 541 TCACGAGCTGGTAATGAATATCTAGGATTAGATCTTCTTAAGGAAGAGTTAATTAAAATG   600
 601 ACCGCTGAAGCCCGCCAGATTCATCCAGAAGGTGTGTTGATTGATGGCGAATTGGTATAC   660
 661 CATGAGCAAGTTAAAAAGGAGCCAGAAGGCCTAGATTTTCTTTTTGATGCTTATCCTGAA   720
 721 AACAGTAAAGCTAAAGAATTCGCCGAAGTAGCTGAATCACGTACTGCTTCTAATGGAATC   780
 781 GCCAATAAATCTTTAAAGGGAACCATTTCTGAAAAAGAAGCACAATGCATGAAGTTTCAG   840
 841 GTCTGGGATTATGTCCCGTTGGTAGAAATATACAGTCTTCCTGCATTTCGTTTGAAATAT   900
 901 GATGTACGTTTTTCTAAACTAGAACAAATGACATCTGGATATGATAAAGTAATTTTAATT   960
 961 GAAAACCAGGTAGTAAATAACCTAGATGAAGCTAAGGTAATTTATAAAAAGTATATTGAC  1020
1021 CAAGGTCTTGAAGGTATTATTCTCAAAAAATATCGATGGATTATGGGAAAATGCTCGTTCA  1080
1081 AAAAATCTTTATAAATTTAAAGAAGTAATTGATGTTGATTTAAAAATTGTAGGAATTTAT  1140
1141 CCTCACCGTAAAGACCCTACTAAAGCGGGTGGATTTATTCTTGAGTCAGAGTGTGGAAAA  1200
1201 ATTAAGGTAAATGCTGGTTCAGGCTTAAAAGATAAAGCCGGTGTAAAATCGCATGAACTT  1260
1261 GACCGTACTCGCATTATGGAAAACCAAAATTATTATATTGGAAAAATTCTAGAGTGCGAA  1320
1321 TGCAACGGTTGGTTAAAATCTGATGGCCGCACTGATTACGTTAAATTATTTCTTCCGATT  1380
1381 GCGATTCGTTTACGTGAAGATAAAACTGAAGCTAATACATTCGAAGATGTATTTGGTGAT  1440
1441 TTTCATGAGGTAACTGGTCTAGGTTCTGGCAGTTCAGGTCATCACCACCATCATCACTAA  1500
```

SEQ ID NO: 24 T4 DNA Ligase CH K470E Protein

```
  1 MILKILNEIASIGSTKQKQAILEKNKDNELLKRVYRLTYSRGLQYYIKKWPKPGIATQSF    60
 61 GMLTLTDMLDFIEFTLATRKLTGNAAIEELTGYITDGKKDDVEVLRRVMMRDLECGASVS   120
```

-continued

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121 IANKVWPGLIPEQPQMLASSYDEKGINKNIKFPAFAQLKADGARCFAEVRGDELDDVRLL     180

181 SRAGNEYLGLDLLKEELIKMTAEARQIHPEGVLIDGELVYHEQVKKEPEGLDFLFDAYPE     240

241 NSKAKEFAEVAESRTASNGIANKSLKGTISEKEAQCMKFQVWDYVPLVEIYSLPAFRLKY     300

301 DVRFSKLEQMTSGYDKVILIENQVVNNLDEAKVIYKKYIDQGLEGIILKNIDGLWENARS     360

361 KNLYKFKEVIDVDLKIVGIYPHRKDPTKAGGFILESECGKIKVNAGSGLKDKAGVKSHEL     420

421 DRTRIMENQNYYIGKILECECNGWLKSDGRTDYVKLFLPIAIRLREDKTEANTFEDVFGD     480

481 FHEVTGLGSGSSGHHHHHH*                                             499
```

The specific methods and compositions described herein are representative of preferred embodiments and are exemplary and not intended as limitations on the scope of the invention. Other objects, aspects, and embodiments will occur to those skilled in the art upon consideration of this specification, and are encompassed within the spirit of the invention as defined by the scope of the claims. It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably may be practiced in the absence of any element or elements, or limitation or limitations, which is not specifically disclosed herein as essential. Thus, for example, in each instance herein, in embodiments or examples of the present invention, any of the terms "comprising", "including", containing", etc. are to be read expansively and without limitation. The methods and processes illustratively described herein suitably may be practiced in differing orders of steps, and that they are not necessarily restricted to the orders of steps indicated herein or in the claims. It is also noted that as used herein and in the appended claims, the singular forms "a," "an," and "the" include plural reference, and the plural include singular forms, unless the context clearly dictates otherwise. Under no circumstances may the patent be interpreted to be limited to the specific examples or embodiments or methods specifically disclosed herein. Under no circumstances may the patent be interpreted to be limited by any statement made by any Examiner or any other official or employee of the Patent and Trademark Office unless such statement is specifically and without qualification or reservation expressly adopted in a responsive writing by Applicants.

The invention has been described broadly and generically herein. Each of the narrower species and subgeneric groupings falling within the generic disclosure also form part of the invention. The terms and expressions that have been employed are used as terms of description and not of limitation, and there is no intent in the use of such terms and expressions to exclude any equivalent of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention as claimed. Thus, it will be understood that although the present invention has been specifically disclosed by preferred embodiments and optional features, modification and variation of the concepts herein disclosed may be resorted to by those skilled in the art, including but not limited to Variant Sequences, and that such modifications and variations are considered to be within the scope of this invention as defined by the appended claims.

SEQUENCE LISTING

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<160> NUMBER OF SEQ ID NOS: 26

<210> SEQ ID NO 1
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
      Bacteriophage T4 sequence

<400> SEQUENCE: 1

atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca      60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct     120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt     180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa     240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat     300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct     360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct     420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct     480
```

```
gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
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<210> SEQ ID NO 2
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
Bacteriophage T4 sequence

<400> SEQUENCE: 2

```
Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175
```

```
Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 3
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 3

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attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180
```

```
ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaaaaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
```

<210> SEQ ID NO 4
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 4

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Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Lys Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125
```

```
Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 5
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 5

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attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120
cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180
ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240
ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300
gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360
attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420
tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480
gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540
tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600
accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660
catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720
aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780
gccaataaat ctttaaaggg aaccatttct aaaaaagaag cacaatgcat gaagtttcag    840
gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900
gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960
gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020
caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080
aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140
cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200
attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260
gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320
tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380
gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440
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<210> SEQ ID NO 6
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 6

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1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80
```

```
Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Lys Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 7
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 7

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atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca     60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattcgt   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
```

<210> SEQ ID NO 8
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 8

```
Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30
```

-continued

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Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Arg Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
```

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    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His


<210> SEQ ID NO 9
<211> LENGTH: 1499
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 9

atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca     60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt caattgattt aaaaattgta ggaatttatc   1140

ctcaccgtaa agaccctact aaagcgggtg gatttattct tgagtcagag tgtggaaaaa   1200

ttaaggtaaa tgctggttca ggcttaaaag ataaagccgg tgtaaaatcg catgaacttg   1260

accgtactcg cattatggaa aaccaaaatt attatattgg aaaaattcta gagtgcgaat   1320

gcaacggttg gttaaaatct gatggccgca ctgattacgt taaattattt cttccgattg   1380

cgattcgttt acgtgaagat aaaactaaag ctaatacatt cgaagatgta tttggtgatt   1440

ttcatgaggt aactggtcta ggttctggca gttcaggtca tcaccaccat catcactaa    1499


<210> SEQ ID NO 10
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 10

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1 5 10 15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
20 25 30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
35 40 45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
50 55 60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65 70 75 80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
85 90 95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
100 105 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
115 120 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
130 135 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145 150 155 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
165 170 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
180 185 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
195 200 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
210 215 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225 230 235 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
245 250 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
260 265 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
275 280 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
290 295 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305 310 315 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
325 330 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
340 345 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
355 360 365

Val Ile Gln Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
370 375 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385 390 395 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys

```
            405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
        420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
    435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
            485                 490                 495

His His His


<210> SEQ ID NO 11
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 11

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attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt cgtgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
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<210> SEQ ID NO 12
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 12

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
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        355                 360                 365

Val Ile Arg Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His


<210> SEQ ID NO 13
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 13

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attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200
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attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcataaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500


<210> SEQ ID NO 14
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 14

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
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305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Lys Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 15
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 15

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atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca     60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960
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gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac 1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca 1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat 1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa 1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt 1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct aaaatgcgaa 1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt 1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat 1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa 1500

<210> SEQ ID NO 16
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 16

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1 5 10 15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
20 25 30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
35 40 45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
50 55 60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65 70 75 80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
85 90 95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
100 105 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
115 120 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
130 135 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145 150 155 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
165 170 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
180 185 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
195 200 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
210 215 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225 230 235 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
245 250 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys

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            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Lys Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His


<210> SEQ ID NO 17
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 17

atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca      60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct     120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt     180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa     240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat     300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct     360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct     420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct     480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta     540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg     600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac     660
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catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgccgt   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500


<210> SEQ ID NO 18
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 18

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
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    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Arg Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His


<210> SEQ ID NO 19
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 19

atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca     60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420
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tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgctgg   1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
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<210> SEQ ID NO 20
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 20

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
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                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Trp Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 21
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 21

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atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca     60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct    120
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cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt    180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa    240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat    300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct    360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct    420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct    480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta    540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg    600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac    660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa    720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc    780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag    840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat    900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt    960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac   1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca   1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat   1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa   1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt   1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa   1320

tgcaacggtt ggttaaaatc tgatggccgc actcgttacg ttaaattatt tcttccgatt   1380

gcgattcgtt tacgtgaaga taaaactaaa gctaatacat tcgaagatgt atttggtgat   1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa   1500
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<210> SEQ ID NO 22
<211> LENGTH: 499
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 22

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
1               5                   10                  15

Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
```

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        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Arg Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Lys Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495

His His His
```

<210> SEQ ID NO 23
<211> LENGTH: 1500
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 23

atgattctta aaattctgaa cgaaatagca tctattggtt caactaaaca gaagcaagca      60

attcttgaaa agaataaaga taatgaattg cttaaacgag tatatcgtct gacttattct     120

cgtgggttac agtattatat caagaaatgg cctaaacctg gtattgctac ccagagtttt     180

ggaatgttga ctcttaccga tatgcttgac ttcattgaat tcacattagc tactcggaaa     240

ttgactggaa atgcagcaat tgaggaatta actggatata tcaccgatgg taaaaaagat     300

gatgttgaag ttttgcgtcg agtgatgatg cgagaccttg aatgtggtgc ttcagtatct     360

attgcaaaca aagtttggcc aggtttaatt cctgaacaac ctcaaatgct cgcaagttct     420

tatgatgaaa aaggcattaa taagaatatc aaatttccag cctttgctca gttaaaagct     480

gatggagctc ggtgttttgc tgaagttaga ggtgatgaat tagatgatgt tcgtctttta     540

tcacgagctg gtaatgaata tctaggatta gatcttctta aggaagagtt aattaaaatg     600

accgctgaag cccgccagat tcatccagaa ggtgtgttga ttgatggcga attggtatac     660

catgagcaag ttaaaaagga gccagaaggc ctagattttc tttttgatgc ttatcctgaa     720

aacagtaaag ctaaagaatt cgccgaagta gctgaatcac gtactgcttc taatggaatc     780

gccaataaat ctttaaaggg aaccatttct gaaaaagaag cacaatgcat gaagtttcag     840

gtctgggatt atgtcccgtt ggtagaaata tacagtcttc ctgcatttcg tttgaaatat     900

gatgtacgtt tttctaaact agaacaaatg acatctggat atgataaagt aattttaatt     960

gaaaaccagg tagtaaataa cctagatgaa gctaaggtaa tttataaaaa gtatattgac    1020

caaggtcttg aaggtattat tctcaaaaat atcgatggat tatgggaaaa tgctcgttca    1080

aaaaatcttt ataaatttaa agaagtaatt gatgttgatt taaaaattgt aggaatttat    1140

cctcaccgta aagaccctac taaagcgggt ggatttattc ttgagtcaga gtgtggaaaa    1200

attaaggtaa atgctggttc aggcttaaaa gataaagccg gtgtaaaatc gcatgaactt    1260

gaccgtactc gcattatgga aaaccaaaat tattatattg gaaaaattct agagtgcgaa    1320

tgcaacggtt ggttaaaatc tgatggccgc actgattacg ttaaattatt tcttccgatt    1380

gcgattcgtt tacgtgaaga taaaactgaa gctaatacat tcgaagatgt atttggtgat    1440

tttcatgagg taactggtct aggttctggc agttcaggtc atcaccacca tcatcactaa    1500


<210> SEQ ID NO 24
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 24

Met Ile Leu Lys Ile Leu Asn Glu Ile Ala Ser Ile Gly Ser Thr Lys
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Gln Lys Gln Ala Ile Leu Glu Lys Asn Lys Asp Asn Glu Leu Leu Lys
            20                  25                  30

Arg Val Tyr Arg Leu Thr Tyr Ser Arg Gly Leu Gln Tyr Tyr Ile Lys
        35                  40                  45

Lys Trp Pro Lys Pro Gly Ile Ala Thr Gln Ser Phe Gly Met Leu Thr
    50                  55                  60

Leu Thr Asp Met Leu Asp Phe Ile Glu Phe Thr Leu Ala Thr Arg Lys
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-continued

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65                  70                  75                  80

Leu Thr Gly Asn Ala Ala Ile Glu Glu Leu Thr Gly Tyr Ile Thr Asp
                85                  90                  95

Gly Lys Lys Asp Asp Val Glu Val Leu Arg Arg Val Met Met Arg Asp
            100                 105                 110

Leu Glu Cys Gly Ala Ser Val Ser Ile Ala Asn Lys Val Trp Pro Gly
        115                 120                 125

Leu Ile Pro Glu Gln Pro Gln Met Leu Ala Ser Ser Tyr Asp Glu Lys
    130                 135                 140

Gly Ile Asn Lys Asn Ile Lys Phe Pro Ala Phe Ala Gln Leu Lys Ala
145                 150                 155                 160

Asp Gly Ala Arg Cys Phe Ala Glu Val Arg Gly Asp Glu Leu Asp Asp
                165                 170                 175

Val Arg Leu Leu Ser Arg Ala Gly Asn Glu Tyr Leu Gly Leu Asp Leu
            180                 185                 190

Leu Lys Glu Glu Leu Ile Lys Met Thr Ala Glu Ala Arg Gln Ile His
        195                 200                 205

Pro Glu Gly Val Leu Ile Asp Gly Glu Leu Val Tyr His Glu Gln Val
    210                 215                 220

Lys Lys Glu Pro Glu Gly Leu Asp Phe Leu Phe Asp Ala Tyr Pro Glu
225                 230                 235                 240

Asn Ser Lys Ala Lys Glu Phe Ala Glu Val Ala Glu Ser Arg Thr Ala
                245                 250                 255

Ser Asn Gly Ile Ala Asn Lys Ser Leu Lys Gly Thr Ile Ser Glu Lys
            260                 265                 270

Glu Ala Gln Cys Met Lys Phe Gln Val Trp Asp Tyr Val Pro Leu Val
        275                 280                 285

Glu Ile Tyr Ser Leu Pro Ala Phe Arg Leu Lys Tyr Asp Val Arg Phe
    290                 295                 300

Ser Lys Leu Glu Gln Met Thr Ser Gly Tyr Asp Lys Val Ile Leu Ile
305                 310                 315                 320

Glu Asn Gln Val Val Asn Asn Leu Asp Glu Ala Lys Val Ile Tyr Lys
                325                 330                 335

Lys Tyr Ile Asp Gln Gly Leu Glu Gly Ile Ile Leu Lys Asn Ile Asp
            340                 345                 350

Gly Leu Trp Glu Asn Ala Arg Ser Lys Asn Leu Tyr Lys Phe Lys Glu
        355                 360                 365

Val Ile Asp Val Asp Leu Lys Ile Val Gly Ile Tyr Pro His Arg Lys
    370                 375                 380

Asp Pro Thr Lys Ala Gly Gly Phe Ile Leu Glu Ser Glu Cys Gly Lys
385                 390                 395                 400

Ile Lys Val Asn Ala Gly Ser Gly Leu Lys Asp Lys Ala Gly Val Lys
                405                 410                 415

Ser His Glu Leu Asp Arg Thr Arg Ile Met Glu Asn Gln Asn Tyr Tyr
            420                 425                 430

Ile Gly Lys Ile Leu Glu Cys Glu Cys Asn Gly Trp Leu Lys Ser Asp
        435                 440                 445

Gly Arg Thr Asp Tyr Val Lys Leu Phe Leu Pro Ile Ala Ile Arg Leu
    450                 455                 460

Arg Glu Asp Lys Thr Glu Ala Asn Thr Phe Glu Asp Val Phe Gly Asp
465                 470                 475                 480

Phe His Glu Val Thr Gly Leu Gly Ser Gly Ser Ser Gly His His His
                485                 490                 495
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His His His


<210> SEQ ID NO 25
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 25

Gly Ser Gly Ser Ser Gly His His His His His His
1               5                   10


<210> SEQ ID NO 26
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      6xHis tag

<400> SEQUENCE: 26

His His His His His His
1               5
```

What is claimed is:

1. A mutant T4 DNA ligase amino acid sequence having only one of the following the amino acid sequences and mutations: E271K in the amino acid sequence of SEQ ID NO: 6, E419K in the amino acid sequence of SEQ ID NO: 14, D452R in the amino acid sequence of SEQ ID NO: 22, and further, wherein each said mutant T4 DNA ligase amino acid sequence does not include the a 6-membered histidine tag at its C-terminus, or a sequence GLGSGSSG preceding the 6-membered histidine tag, and wherein said mutant T4 DNA ligase has greater ligase activity at lower concentrations as compared to a wild type T4 ligase without said mutations.

2. A mutant T4 DNA ligase amino acid sequence of claim **1** having the mutation E271K in the amino acid sequence of SEQ ID NO: 6.

3. A mutant T4 DNA ligase amino acid sequence of claim **1** having the mutation E419K in the amino acid sequence of SEQ ID NO: 14.

4. A mutant T4 DNA ligase amino acid sequence of claim **1** having the mutation D452R in the amino acid sequence of SEQ ID NO: 22.

5. A polynucleotide encoding the amino acid sequence of the mutant T4 DNA ligase of claim **1**.

6. A vector comprising the polynucleotide of claim **5**.

7. A cell transformed with the vector of claim **6**.

8. A process of conducting polynucleotide ligation between different polynucleotides by ligating the 5' and 3' ends of polynucleotides to generate a circular polynucleotide, wherein the polynucleotides have blunt ends or cohesive ends, comprising:

providing a ligation mixture including the polynucleotides to be ligated and the mutant T4 DNA ligase of claim **1**; and placing said ligation mixture at temperatures wherein ligation takes place, thereby generating said circular polynucleotide.

9. A process of conducting polynucleotide ligation between different polynucleotides by ligating the 5' and 3' ends of polynucleotides to generate a circular polynucleotide, wherein the polynucleotides have blunt ends or cohesive ends, comprising:

providing a ligation reaction mixture including a buffer, the polynucleotides to be ligated and the mutant T4 DNA ligase of claim **1**; and placing said ligation reaction mixture under temperature conditions suitable for ligation, thereby generating said circular polynucleotide.

10. The process of claim **9** wherein the ligation reaction mixture comprises Tris-HCl, MgCl2, ATP, dithiothreitol and water.

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